

Avenanthramides inhibit proliferation of human colon cancer cell lines in vitro.

A high intake of whole grain foods is associated with reduced risk of colon cancer, but the mechanism underlying this protection has yet to be elucidated. Chronic inflammation and associated cyclooxygenase-2 (COX-2) expression in the colon epithelium are causally related to epithelial carcinogenesis, proliferation, and tumor growth. We examined the effect of avenanthramides (Avns), unique polyphenols from oats with anti-inflammatory properties, on COX-2 expression in macrophages, colon cancer cell lines, and on proliferation of human colon cancer cell lines. We found that Avns-enriched extract of oats (AvExO) had no effect on COX-2 expression, but it did inhibit COX enzyme activity and prostaglandin E(2) (PGE(2)) production in lipopolysaccharide-stimulated mouse peritoneal macrophages. Avns (AvExO, Avn-C, and the methylated form of Avn-C (CH3-Avn-C)) significantly inhibited cell proliferation of both COX-2-positive HT29, Caco-2, and LS174T, and COX-2-negative HCT116 human colon cancer cell lines, CH3-Avn-C being the most potent. However, Avns had no effect on COX-2 expression and PGE(2) production in Caco-2 and HT29 colon cancer cells. These results indicate that the inhibitory effect of Avns on colon cancer cell proliferation may be independent of COX-2 expression and PGE(2) production. Thus, Avns might reduce colon cancer risk through inhibition of macrophage PGE(2) production and non-COX-related antiproliferative effects in colon cancer cells. Interestingly, Avns had no effect on cell viability of confluence-induced differentiated Caco-2 cells, which display the characteristics of normal colonic epithelial cells. Our results suggest that the consumption of oats and oat bran may reduce the risk of colon cancer not only because of their high fiber content but also due to Avns, which attenuate proliferation of colonic cancer cells.